

Multi-Agent Reinforcement Learning for Smart Grid Management: A Technical Survey

TL;DR

Smart grids are intrinsically multi-agent, partially observable, and non-stationary, which makes MARL a natural fit for coordinating distributed energy resources (DERs), EVs, storage, and microgrids; the strongest published results are in distribution-level volt/VAR control, EV charging, demand response, and transmission topology control (the L2RPN competitions), but the field is dominated by simulation studies with a large, honestly-acknowledged sim-to-real gap.

The dominant algorithmic paradigm is Centralized Training with Decentralized Execution (MADDPG, MAPPO, QMIX, COMA, VDN, QPLEX), increasingly combined with graph neural networks that exploit grid topology and with safe/constrained-RL layers (Lagrangian, safety projection, control-barrier functions) required to respect voltage/thermal/frequency limits.

As of 2026 there is no documented autonomous closed-loop MARL controller running a major grid; real activity is at the advisory/decision-support, pilot, and hardware-in-the-loop stage (RTE France's L2RPN and the AI4REALNET project, national-lab work at NREL/PNNL/ANL/Sandia, EPRI research), and much commercial "AI for DER" is optimization or forecasting rather than MARL.

Key Findings

Genuine value vs. classical methods. MARL adds value where the problem is high-dimensional, stochastic, fast, decentralized, and privacy-constrained — e.g., real-time volt/VAR control with thousands of smart inverters, coordinating large EV/prosumer populations, and combinatorial transmission topology control where AC-OPF and MILP become intractable in real time. Classical methods (AC-OPF, MPC, ADMM, stochastic/robust programming, MILP unit commitment) remain superior where models are accurate, constraints are hard and must be provably satisfied, and offline compute is available. The honest consensus is that MARL is a fast approximate policy / decision-support layer, not a replacement for the optimization and protection stack.

CTDE is the workhorse. MADDPG and MAPPO (continuous/mixed control) and value-decomposition methods (VDN, QMIX, QTRAN, QPLEX) plus COMA (credit assignment) are the most common backbones in power-systems papers. GNN encoders are increasingly bolted on to generalize across topology and N-1 contingencies.

Safety is the deployment bottleneck. Because utilities will not deploy black-box controllers that can violate voltage/thermal/frequency limits, a large sub-literature adds constrained-MDP/Lagrangian methods, safety projection layers, shielding, and control-barrier/Lyapunov functions. These reduce but do not eliminate the verification gap.

Benchmarks exist but standardization is weak. Grid2Op/L2RPN (transmission topology) and CityLearn (building/district demand response) are the two most mature RL-specific benchmarks; PowerGridworld, PowerGym, GridLearn, and simulators like pandapower, OpenDSS, PyPSA, GridLAB-D, ANDES, and RLGC are widely used. Reproducibility and standardized evaluation remain poor.

The literature is survey-rich and hype-aware. Multiple 2020–2024 surveys in IEEE Transactions on Smart Grid, IEEE Transactions on Power Systems, Applied Energy, and Renewable & Sustainable Energy Reviews

now explicitly warn about the simulation-to-deployment gap, reward hacking, and lack of guarantees.

Details

1. Problem framing and motivation

Why grids are multi-agent and partially observable. Modern distribution and transmission systems comprise many autonomous or semi-autonomous decision-makers: distributed generators (rooftop PV, wind), grid-following/grid-forming inverters, battery energy storage, EV chargers, controllable loads, prosumers, aggregators/VPPs, and networked microgrids. Each observes only local measurements (its bus voltage, local power flows), communication is bandwidth- and latency-limited, ownership is fragmented (privacy matters), and the environment is non-stationary (weather, prices, demand, topology changes). This maps cleanly onto decentralized sequential decision-making under uncertainty.

Why single-agent RL and classical optimization fall short.

Single-agent RL requires a central controller with full observability and a joint action space that grows exponentially with the number of devices — infeasible for thousands of inverters/EVs, and brittle to communication loss.

AC Optimal Power Flow (OPF) is a non-convex program; solving it in real time for large networks with discrete decisions is hard, and it needs an accurate model.

Model Predictive Control (MPC) needs a good forward model and can be computationally heavy for fast timescales and large fleets.

ADMM / distributed optimization gives decentralization and (under convexity) convergence guarantees, but struggles with non-convexity, discrete decisions, and requires iterative communication rounds each control step.

Stochastic/robust programming handles uncertainty but scales poorly and needs distributional assumptions.

MARL's value proposition: amortize the optimization into a fast decentralized policy learned offline, handle uncertainty via experience rather than explicit models, and act with local observations at millisecond-second timescales.

Where classical methods remain superior: hard-constraint guarantees, protection and stability-critical control, planning/unit-commitment with provable optimality gaps, and any setting where the physical model is trustworthy and compute is available offline.

Formalizations.

Dec-POMDP (decentralized partially observable MDP) — the canonical model for cooperative grid control with local observations.

Markov / stochastic games — for competitive or mixed settings (market bidding, P2P trading).

Networked MARL — agents on a communication graph, sharing information only with neighbors; consensus-based value/policy learning.

Mean-field games / mean-field MARL — for very large homogeneous populations (thousands of EVs, households), replacing pairwise interactions with an interaction with the population's mean field.

Constrained MDPs (CMDP) and safe RL — encoding voltage/thermal/frequency limits as constraints rather than soft penalties.

2. Algorithmic landscape

CTDE methods and power-system use.

MADDPG — Lowe, Wu, Tamar, Harb, Abbeel & Mordatch, "Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments," NeurIPS 2017 (arXiv:1706.02275); multi-agent DDPG with centralized critics, heavily used for continuous inverter/reactive-power and storage control.

MAPPO (Yu et al., 2021/2022) — multi-agent PPO; strong, stable baseline increasingly favored for volt/VAR and EV problems.

COMA (Foerster et al., AAAI 2018) — counterfactual baseline for credit assignment.

VDN (Sunehag et al., 2018) and **QMIX** (Rashid et al., ICML 2018) — value decomposition for cooperative discrete control; QMIX is common in demand-response and discrete switching tasks.

QTRAN (Son et al., ICML 2019) and **QPLEX** (Wang et al., ICLR 2021) — more expressive value factorization relaxing QMIX's monotonicity constraint. These are typically applied by partitioning the grid into zones/agents (per-substation, per-feeder, per-microgrid).

Fully decentralized / networked MARL. Consensus-based actor-critic (e.g., Zhang et al., ICML 2018, "Fully Decentralized MARL with Networked Agents") underpins schemes where agents exchange parameters/value estimates only with neighbors — attractive for privacy and communication limits.

Mean-field MARL. Mean-field RL (Yang et al., ICML 2018) is applied to large EV fleets and household populations, where each agent responds to an aggregate (e.g., aggregate load or price signal).

Safe / constrained MARL. Approaches include: Lagrangian/primal-dual CMDP methods; safety projection layers that project actions onto a feasible set (e.g., using a linearized power-flow model); shielding; control-barrier-function (CBF) and Lyapunov-based methods that provide stability/invariance guarantees for voltage and frequency control. This is one of the most active areas because it directly addresses the deployment blocker.

GNN + RL and physics-informed RL. Graph neural networks encode buses as nodes and lines as edges, giving permutation invariance and better generalization across topology changes and N-1 contingencies — especially important in L2RPN-style topology control. Physics-informed RL embeds power-flow equations or constraints into the policy/critic or reward.

Offline / batch MARL. Because online exploration on a live grid is unsafe, offline RL (learning from logged operational data) is of high interest, though offline MARL is immature and suffers from distributional shift and coordinated extrapolation error.

Hierarchical MARL and 2023–2026 trends. Hierarchical schemes separate slow (setpoint/dispatch) from fast (inverter) timescales. Recent (2023–2026) directions include transformers/Decision-Transformer-style sequence models, foundation models for power systems, and LLM agents as reasoning/explanation layers or planners for grid operation. This LLM/foundation-model line is nascent and largely position-paper/prototype stage; claims should be treated cautiously.

Table 1 — Representative MARL algorithms and grid use.

Algorithm	Type	Paradigm	Typical grid use	Notes/limitations
MADDPG	Actor-critic, continuous	CTDE	Inverter/reactive power, storage	Scales poorly with many agents
MAPPO	On-policy actor-critic	CTDE	Volt/VAR, EV, DR	Stable, sample-hungry
COMA	Actor-critic	CTDE	Credit assignment tasks	Discrete actions
VDN/QMIX	Value decomposition	CTDE	Discrete DR, switching	Monotonicity limits expressiveness
QTRAN/QPLEX	Value decomposition	CTDE	Cooperative control	More expressive, harder to train
Networked AC	Consensus actor-critic	Decentralized	Privacy-limited coordination	Communication overhead
Mean-field MARL	Population	CTDE/decentralized	EV fleets, households	Homogeneity assumption
Safe/constrained MARL	CMDP/CBF/Lyapunov	CTDE	Voltage/frequency limits	Guarantees often approximate
GNN-MARL	GNN encoder + RL	CTDE	Topology control, OPF	Better generalization

3. Application areas

Voltage/VAR control & reactive power — the most mature MARL application in distribution networks; smart PV inverters as agents adjusting reactive power to keep voltages in band (typically $\pm 5\%$ of nominal per ANSI C84.1-2020 Range A, i.e., 95%–105%, or 114–126 V on a 120 V base). MADDPG/MAPPO with safety layers dominate; reported results show voltage-violation reduction and loss reduction on IEEE 33/123-bus feeders in simulation.

Frequency regulation / AGC — MARL for load-frequency control and fast frequency response, made harder by falling inertia in inverter-dominated grids; stability guarantees are the key concern, motivating Lyapunov/CBF approaches.

Economic dispatch, unit commitment, AC-OPF — RL used as a fast approximate solver or warm-start; classical MILP/convex methods remain the reference for optimality and constraints.

Demand response / DSM / HEMS — MARL coordinates flexible loads and home energy management; strong benchmark presence via CityLearn.

P2P energy trading / local markets / transactive energy — modeled as Markov games; MARL learns bidding/trading strategies; raises tacit-collusion concerns.

Microgrids & networked microgrids — coordination, islanding, black start, resource sharing; MARL for decentralized dispatch and resilience.

EV charging & V2G — large-population coordination (mean-field and CTDE MARL) to flatten load, respect transformer limits, and provide grid services.

Energy storage / battery arbitrage — scheduling against prices and grid constraints; often combined with DR and PV.

Network reconfiguration, service restoration, resilience — topology control after faults/extreme events; the L2RPN task is the flagship transmission version.

Cybersecurity — MARL for coordinated attack detection/mitigation, plus a growing adversarial-ML literature showing MARL grid controllers are vulnerable to observation/action perturbations and adversarial agents (the L2RPN "Robustness" track built in an adversarial opponent).

Wholesale market bidding — RL agents learning strategic bids; research flags the risk of learned tacit collusion among independent RL bidders.

4. Environments, benchmarks, datasets, tools

Grid2Op / L2RPN — RTE France's platform (lead developer Benjamin Donnot) and the "Learning to Run a Power Network" competition series for transmission topology control (busbar switching, line switching, redispatch, storage). Editions run at major venues from 2019 onward (IEEE WCCI 2019, NeurIPS 2020 with Robustness and Adaptability tracks, ICAPS 2021 "Trust", WCCI 2022 energy management), using grids based on IEEE 14-bus (early) and ~36-substation subsets of the IEEE 118-bus system (later). Winning recipes: imitation learning + PPO with simulation-guided action-space reduction, curriculum learning, and the Semi-Markov Afterstate Actor-Critic (SMAAC, from Yoon, Kim, Ok & Yun, "Winning the L2RPN Challenge: Power Grid Management via Semi-Markov Afterstate Actor-Critic," ICLR 2021, KAIST — which won the L2RPN WCCI 2020 and NeurIPS 2020 tracks). A key competition survey is Marot, Donnot et al., "Learning to run a power network challenge," Electric Power Systems Research, ~2020.

CityLearn — introduced by Vázquez-Canteli, Kämpf, Henze & Nagy, "CityLearn v1.0: An OpenAI Gym Environment for Demand Response with Deep Reinforcement Learning," BuildSys 2019, developed at the University of Texas at Austin (Intelligent Environments Laboratory); hosted the CityLearn Challenge editions and is widely used for cooperative MARL in energy communities.

Other RL environments — PowerGridworld (NREL), GridLearn, PowerGym (volt/VAR on OpenDSS feeders), RLGC (RL for grid control, PNNL, used for emergency voltage/frequency control).

Simulators/tools — pandapower, PyPSA, OpenDSS, GridLAB-D, MATPOWER, GridPACK (PNNL), ANDES/OpenModelica (dynamics).

Standard test feeders — IEEE 13/33/34/123/141-bus and CIGRE benchmarks for distribution; IEEE 14/30/57/118/300-bus for transmission.

Reproducibility problems — inconsistent feeders, load/PV profiles, reward definitions, and evaluation metrics; few papers release code/data; sim-to-real gap poorly quantified.

Table 2 — Major benchmark environments.

Environment	Domain	Backend	Task focus	Maintainer
Grid2Op / L2RPN	Transmission	Custom power flow	Topology control, redispatch	RTE France
CityLearn	Buildings/district	EnergyPlus-derived data	DR, load shaping	UT Austin / community
PowerGym	Distribution	OpenDSS	Volt/VAR	community
PowerGridworld	Distribution/DER	pandapower/custom	Multi-agent DER	NREL
GridLearn	Distribution	pandapower/CityLearn	Volt/VAR, multi-agent	community
RLGC	Transmission dynamics	InterPSS/dynamics	Emergency control	PNNL

5. Key technical challenges

Non-stationarity & credit assignment — each learning agent changes others' environment; assigning global reward to individual actions is hard (COMA, value decomposition address this partially).

Scalability — to thousands of agents; mean-field and parameter-sharing help but homogeneity assumptions bite.

Sample efficiency & simulation cost — high-fidelity power-flow/dynamics simulation is expensive; offline RL and model-based methods are being explored.

Safety & constraint satisfaction — the central deployment blocker; guarantees are usually approximate.

Partial observability, communication, latency, privacy — realistic comms constraints are often ignored in papers.

Sim-to-real & robustness — topology changes, N-1 contingencies, and model mismatch degrade learned policies.

Reward design & multi-objective trade-offs — cost vs. emissions vs. reliability vs. fairness; reward hacking is common.

Interpretability, verification, regulation — NERC reliability standards, IEEE Std 1547-2018 ("IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces"), and grid codes create high bars for black-box controllers.

6. Real-world deployments and industry activity

RTE France runs L2RPN and participates in AI4REALNET (AI for REAL-world NETwork operation), a Horizon Europe project launched October 2023 with a €5.98M EU contribution, coordinated by INESC TEC and involving RTE (CORDIS Grant agreement ID: 101119527), precisely to move topology-control RL toward operator decision-support; usage is advisory/research, not autonomous control.

National Grid ESO (UK) uses ML mainly for forecasting/balancing; no documented RL/MARL closed-loop control.

EPRI runs AI/ML-for-grid research programs (volt/VAR, DER, resilience) at pilot/research stage.

US national labs — NREL (DER/building/EV RL, HIL testbeds), PNNL (RLGC, volt/VAR, resilience, VOLTTRON agent middleware, GridPACK), ANL (markets, microgrids), Sandia (microgrid control,

resilience, security). Mostly simulation/HIL.

Adjacent industry ML — Google/DeepMind's data-center cooling and wind-farm output forecasting (2019) are frequently cited but are generation/forecasting, not grid control.

Startups/vendors — AutoGrid Systems (VPP/DER management), whose acquisition by Schneider Electric was announced May 9, 2022, with Schneider describing AutoGrid as "a leader in Distributed Energy Resource Management Systems (DERMS)"; Gridmatic (energy trading ML); and DER-flexibility platforms (Leap, Voltus, Piclo, Octopus/Kraken). Many use optimization/forecasting rather than MARL, and marketing often overstates "AI."

Honest assessment — there is a wide gap between published simulation results and deployed systems; no autonomous MARL grid controller is in production as of 2026.

7. State of the field and future directions

Surveys/taxonomies — several 2020–2024 reviews in IEEE Transactions on Smart Grid, IEEE Transactions on Power Systems, Applied Energy, and Renewable & Sustainable Energy Reviews taxonomize by application, algorithm, and formalization, and increasingly emphasize safety and reproducibility.

Emerging trends — foundation models for power systems, LLM agents for operator support/explainability, digital twins for training, federated MARL for privacy, and hybrid physics-ML (physics-informed and safe RL).

Open problems — provable safety/verification, offline MARL, robust sim-to-real transfer, standardized benchmarks/metrics, scalable credit assignment, and market-manipulation/collusion analysis.

Groups/venues — RTE France, KAIST, TU Delft, national labs (NREL/PNNL/ANL/Sandia), and academic groups publishing at IEEE Trans. Smart Grid, IEEE Trans. Power Systems, Applied Energy, PSCC, IEEE PES General Meeting, and NeurIPS/ICML/ICLR plus the "Tackling Climate Change with ML" workshop.

Recommendations

If writing a literature review: organize by (a) formalization, (b) algorithm family (CTDE/networked/mean-field/safe/GNN), and (c) application, and dedicate a section to the sim-to-real gap and reproducibility; anchor on Grid2Op/L2RPN and CityLearn as the reproducible benchmarks.

If planning a research project: target one of the high-leverage open problems — safe/constrained MARL with verifiable guarantees, offline MARL from utility data, or standardized benchmarking with sim-to-real evaluation. Use pandapower/OpenDSS + PowerGridworld/PowerGym for distribution work and Grid2Op for transmission.

If evaluating the field for deployment: treat MARL as decision-support/advisory today; require a safety layer (projection/CBF) and hardware-in-the-loop validation before any pilot; benchmark against MPC/ADMM/OPF baselines, which are still the reference.

Thresholds that change the recommendation: a peer-reviewed field pilot demonstrating constraint satisfaction under N-1 on a real feeder, or a standardized benchmark with reported sim-to-real transfer, would justify moving from advisory to supervised-autonomy pilots.

Caveats

Live web search was unavailable during this research; the report is compiled from domain knowledge, with several specifics corroborated during an enrichment pass. Remaining specifics — exact competition

editions, full winning-team rosters, precise grid sizes, and any numerical performance figures — should be verified against primary sources (l2rpn.chalearn.org, the Grid2Op GitHub, and the cited papers) before publication.

Several 2024–2026 claims (LLM agents, foundation models) reflect an emerging, fast-moving area where specific papers could not be confirmed; treat as leads, not confirmed citations.

Vendor "AI" claims are frequently marketing; distinguish MARL from optimization/forecasting.